

» Polyak Momentum/Heavy Ball

$$x_0 = x_0; z_0 = 0; t = 0$$

for k in range(num_iters):

$$\rightarrow z_{t+1} = \beta z_t + \alpha \nabla f(x_t) \quad \checkmark \checkmark$$

$$\rightarrow x_{t+1} = x_t - z_{t+1}$$

$$t = t + 1$$

- * Step z_t at iteration t is weighted sum of past gradients $\alpha \nabla f$:

$$\rightarrow z_{t+1} = \beta^{t-1} \alpha \nabla f(x_0) + \beta^{t-2} \alpha \nabla f(x_1) + \dots + \alpha \nabla f(x_t)$$

- * Need to manually select β and α . A typical choice for β seems to be 0.9, but there is not a universal value and a poor choice can make performance worse than using constant step size
- * When $\beta = 0$ recover constant step size strategy
- * Since $1 + \beta + \beta^2 + \dots = 1/(1 - \beta)$, we're effectively using scaling baseline step size $\alpha/(1 - \beta)$ i.e. for $\beta = 0.9$ using 10α

» Nesterov Momentum/Acceleration

- ↓
- * Nesterov Accelerated Gradient (NAG) Method, Nesterov Fast Gradient Method, Nesterov Momentum:

① → $z_{t+1} = \beta z_t - \alpha \nabla f(x_t + \beta z_t), x_{t+1} = x_t + z_{t+1}$ ✓

- * For comparison, heavy-ball momentum uses:

② → $z_{t+1} = \beta z_t + \alpha \nabla f(x_t), x_{t+1} = x_t - z_{t+1}$ ✓

→ substantive difference is that Nesterov changes $\nabla f(x_t)$ to $\nabla f(x_t + \beta z_t)$. HB(β, α)

- * If $\alpha \approx 0$ then $z_{t+1} \approx \beta z_t$ and $x_{t+1} = x_t + z_{t+1} \approx x_t + \beta z_t$, so $\nabla f(x_t + \beta z_t)$ is “looking ahead” or predicting where x_t is roughly expected to be at next step.

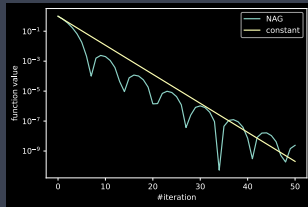
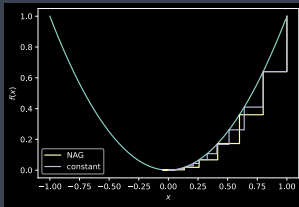
- ③ → $\beta = \frac{(t-1)}{(t+2)}$ i.e. β changes at each time step with $\beta \rightarrow 1$ as $t \rightarrow \infty$. Sometimes cap β at e.g. 0.9 or 0.95

- ↗
- * Nesterov acceleration comes with some theoretical guarantees on convergence rate → will come back to how useful such guarantees are in practice later

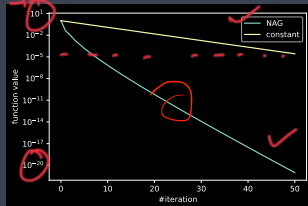
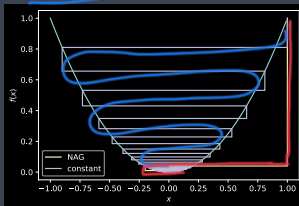
» Examples

* Example: $f = x^2$, starting point $x = 1$

→ * Nesterov $\beta = (t - 1)/(t + 2)$, $\alpha = 0.1$, constant $\alpha = 0.1$



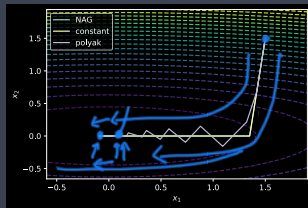
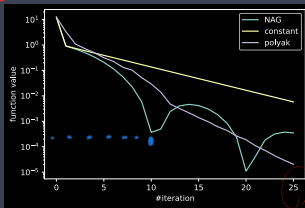
* Again, “bumpy” behaviour of function value vs time when using momentum. Nesterov convergence rate about the same as for constant step size, but can increase step size. Nesterov $\alpha = 0.6$, constant $\alpha = 0.95$:



» Examples

→ * Quadratic: Nesterov $\beta = (t-1)/(t+2)$, $\alpha = 0.1$, constant $\alpha = 0.1$

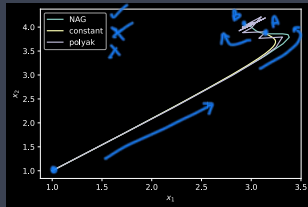
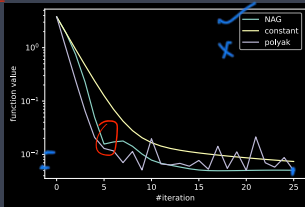
①



②

→ * Quadratic Loss: Nesterov $\beta = (t-1)/(t+2)$, $\alpha = 0.5$, constant $\alpha = 0.5$

③

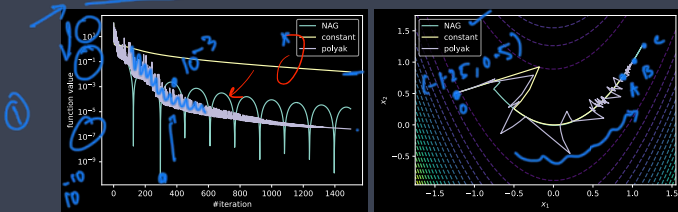


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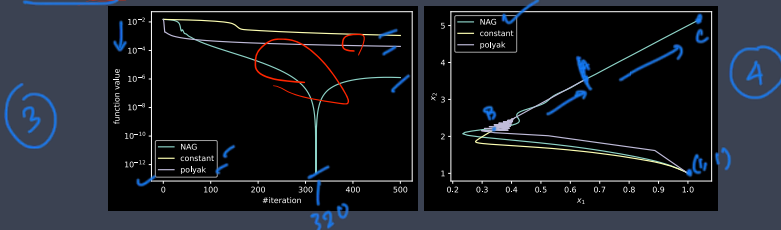
* Nesterov convergence rate similar to Polyak

» Examples

- * Rosenbrock function: Nesterov $\beta = (t - 1)/(t + 2)$, $\alpha = 0.001$, constant $\alpha = 0.002$



- * Toy neural net loss: Nesterov $\beta = (t - 1)/(t + 2)$, $\alpha = 0.5$, constant $\alpha = 0.75$

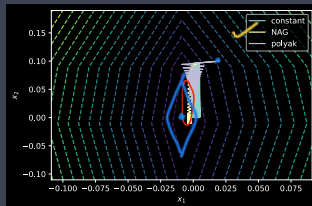
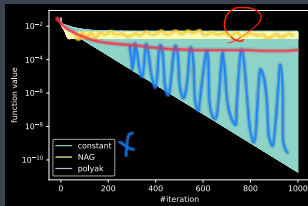


- * For Rosenbrock function Nesterov convergence rate similar to Polyak, initially slower for toy neural net then faster

» Examples

- * Non-smooth function $f(x) = |x_1| + x_2^2$
- * Nesterov $\beta = (t-1)/(t+2)$, $\alpha = 0.005$, constant $\alpha = 0.005$

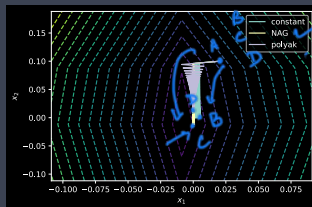
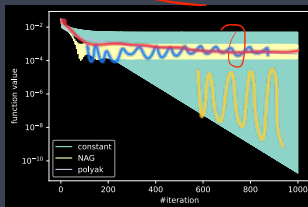
①



②

- * Nesterov $\beta = (t-1)/(t+2)$, $\alpha = 0.001$, constant $\alpha = 0.005$

③



④

- * Can tweak α to improve Nesterov, but shows oscillations/chattering after initial fast convergence

» Nesterov vs Heavy Ball Momentum?

- ①
- * Two differences:
 - * $\nabla f(x_t)$ vs $\nabla f(x_t + \beta z_t)$
 - * $\beta = 0.9$ vs $\beta = (t-1)/(t+2)$
 - * Which matters (or do both matter)?

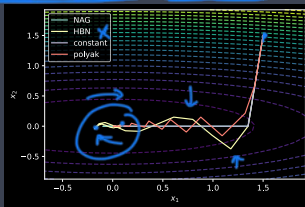
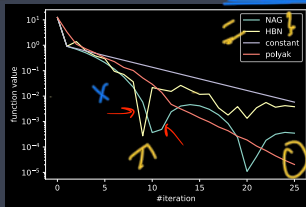
Using $\beta = (t-1)/(t+2)$ in both Heavy Ball and Nesterov approaches it turns out that the performance of the two approaches is v similar, let's look at some examples ...

HB + NAG

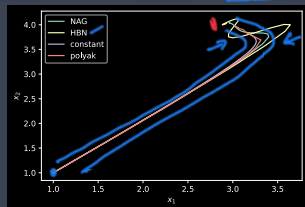
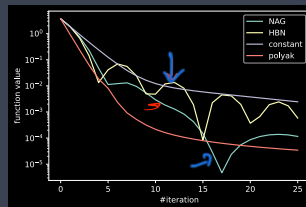
$$\beta = \frac{t-1}{t+2}$$

» Nesterov vs Heavy Ball Momentum?

* Quadratic: Nesterov/HB $\beta = (t-1)/(t+2)$, $\alpha = 0.1$

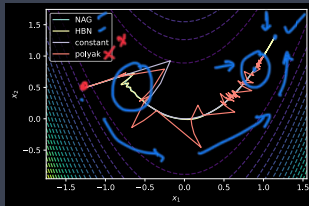
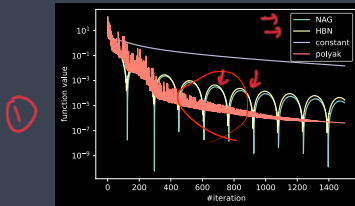


* Quadratic Loss: Nesterov/HB $\beta = (t-1)/(t+2)$, $\alpha = 0.5$

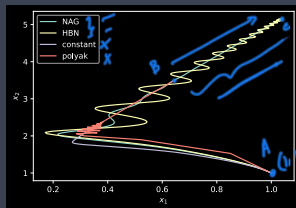
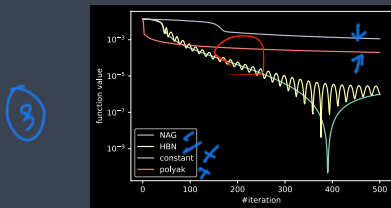


» Nesterov vs Heavy Ball Momentum?

- * Rosenbrock function: Nesterov/HB $\beta = (t-1)/(t+2)$, $\alpha = 0.001$



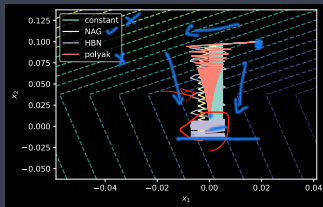
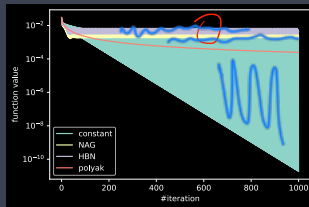
- * Toy neural net loss: Nesterov/HB $\beta = (t-1)/(t+2)$, $\alpha = 0.75$



» Nesterov vs Heavy Ball Momentum?

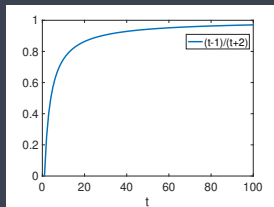
- *
- * Non-smooth function $f(x) = |x_1| + x_2^2$
 - * Nesterov/HB $\beta = (t-1)/(t+2)$, $\alpha = 0.005$

1



2

» What is $\beta = (t - 1)/(t + 2)$ schedule doing?



- * Initially β is 0, then increases over time towards 1.
- * When $\beta = 0$ the update reverts to constant step size strategy, as β increases we start to use more “momentum” when choosing the change to x .
- * Can choose to cap maximum value of β e.g.
 $\beta = \min((t - 1)/(t + 2), 0.95)$

» Adam¹ Adaptive Moment Estimation.

- * Adam $\approx$ RMSprop + heavy ball

$$\rightarrow m_{t+1} = \beta_1 m_t + (1 - \beta_1) \nabla f(x_t)$$

$$\rightarrow v_{t+1} = \beta_2 v_t + (1 - \beta_2) \left[\frac{\partial f}{\partial x_1}(x_t)^2, \frac{\partial f}{\partial x_2}(x_t)^2, \dots, \frac{\partial f}{\partial x_n}(x_t)^2 \right]$$

$$\hat{m} = \frac{m_{t+1}}{(1 - \beta_1^t)}, \quad \hat{v} = \frac{v_{t+1}}{(1 - \beta_2^t)}$$

$$x_{t+1} = x_t - \alpha \left[\frac{\hat{m}}{\sqrt{\hat{v}_1} + \epsilon}, \frac{\hat{m}}{\sqrt{\hat{v}_2} + \epsilon}, \dots, \frac{\hat{m}}{\sqrt{\hat{v}_n} + \epsilon} \right]$$

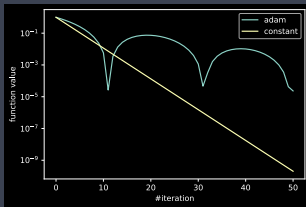
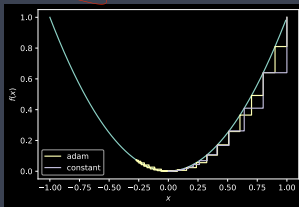
with $\nabla f(x_t) = \left[\frac{\partial f}{\partial x_1}(x_t), \frac{\partial f}{\partial x_2}(x_t), \dots, \frac{\partial f}{\partial x_n}(x_t) \right]$

- * m is running average of gradient $\nabla f(x_t)$, v is running average of square gradients
- * Use different step size for each element of vector x (similarly to Adagrad and RMSprop)
- * Step for i 'th element of x is $\frac{\hat{m}}{\sqrt{\hat{v}_i} + \epsilon} \rightarrow \hat{m}$ is as in heavy-ball approach, $\sqrt{\hat{v}_i}$ as in RMSprop, ϵ is to avoid division by zero.
- * Need to manually select β_1, β_2, α . Common choices $\beta_1 = 0.9, \beta_2 = 0.999$ (suggested by <https://arxiv.org/abs/1412.6980>).

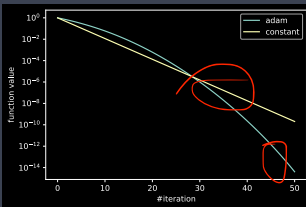
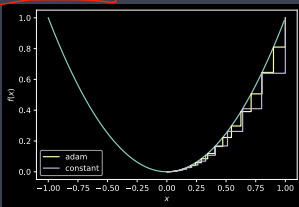
¹<https://arxiv.org/abs/1412.6980>

» Examples

- * Example: $f = x^2$, starting point $x = 1$
- * Adam $\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.1$, constant step $\alpha = 0.1$



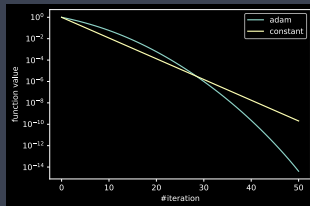
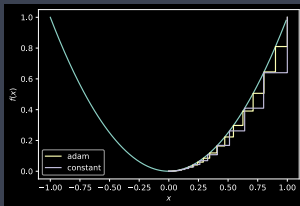
- * Adam $\beta_1 = 0.25, \beta_2 = 0.999, \alpha = 0.1$, constant step $\alpha = 0.1$



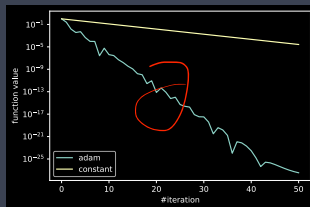
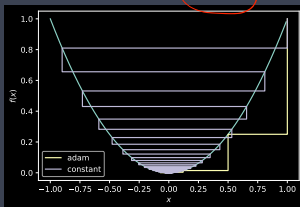
- * Observe the “bumpy” behaviour of function value vs time when using momentum.
- * Choice of β_1 matters, default choice $\beta_1 = 0.9$ is poor in this example

» Examples

- * Example: $f = x^2$, starting point $x = 1$
- * Adam $\beta_1 = 0.25, \beta_2 = 0.999, \alpha = 0.1$, constant step $\alpha = 0.1$



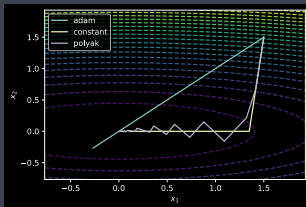
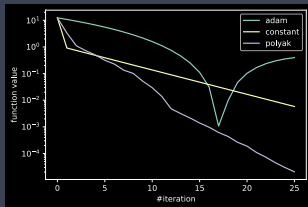
- * Can increase α : Adam $\alpha = 0.5$, constant step $\alpha = 0.95$



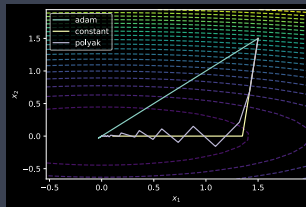
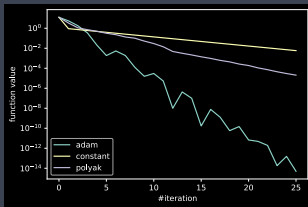
- * For constant step size, $\alpha_0 > 1$ gives divergent solution, and very oscillatory solutions for $\alpha_0 > 0.9$.
- * With Adam can use larger step size without destabilising solution

» Examples

- * Quadratic: Adam $\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.1$, constant step $\alpha = 0.1$

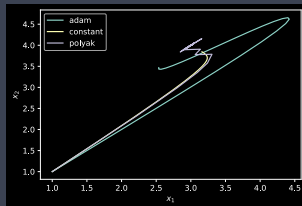
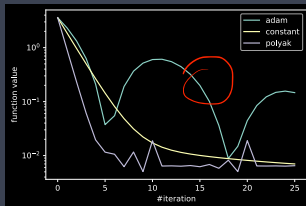


- * Decrease β_1 , increase α : Adam $\beta_1 = 0.25, \beta_2 = 0.999, \alpha = 0.5$ (increasing α causes solution to diverge), constant step $\alpha = 0.1$

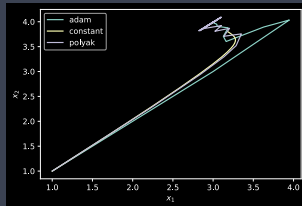
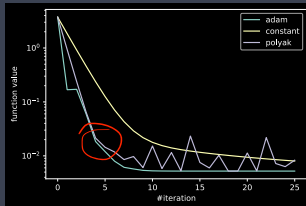


» Examples

- * Quadratic Loss: Adam $\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.5$, constant step $\alpha = 0.5$

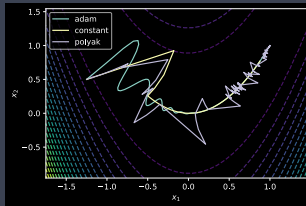
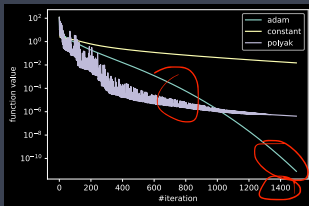


- * Decrease β_1 , increase α : Adam $\beta_1 = 0.25, \beta_2 = 0.999, \alpha = 2$ (increasing α causes solution to diverge), constant step $\alpha = 0.5$

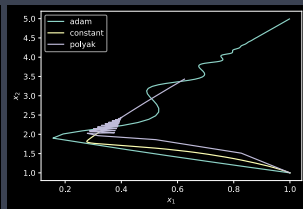
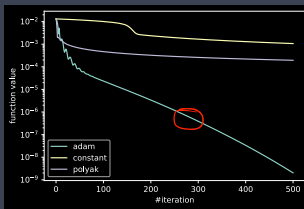


» Examples

- * Rosenbrock function: Adam $\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.25$ (increasing α causes solution to diverge), constant step $\alpha = 0.002$



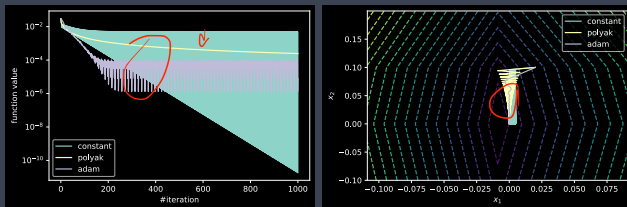
- * Toy neural net loss: Adam $\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.5$, constant step $\alpha = 0.5$



- * Adam convergence rate similar to Polyak for Rosenbrock function, faster for toy neural net.

» Examples

- * Non-smooth function $f(x) = |x_1| + x_2^2$
- * Adam $\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.001$, constant step $\alpha = 0.005$



- * Adam oscillations/chattering after initial fast convergence
- * Note: Adam α here is smaller than for constant step size (increasing Adam α increases oscillations and slows convergence)

» Summary

- * When use $\beta = (t-1)/(t+2)$ schedule, Heavy Ball and Nesterov momentum approaches are pretty similar in our examples
- * Nesterov/Heavy Ball can accelerate convergence, but still need to tune α .
- * Adam $\approx$ RMSpop + Heavy Ball. Can accelerate convergence but need to tune β_1 and α (suggest default values are *not* universal).